```
Dataset used     : glass.csv
Columns          : ['RI', 'Na', 'Mg', 'Al', 'Si', 'K', 'Ca', 'Ba', 'Fe', 'Type']
        RI     Na    Mg    Al     Si     K    Ca   Ba   Fe  Type
0  1.52101  13.64  4.49  1.10  71.78  0.06  8.75  0.0  0.0     1
1  1.51761  13.89  3.60  1.36  72.73  0.48  7.83  0.0  0.0     1
2  1.51618  13.53  3.55  1.54  72.99  0.39  7.78  0.0  0.0     1
3  1.51766  13.21  3.69  1.29  72.61  0.57  8.22  0.0  0.0     1
4  1.51742  13.27  3.62  1.24  73.08  0.55  8.07  0.0  0.0     1

n (rows/examples) : 214
d (features)     : 9

X_train shape    : (149, 9)
X_test  shape    : (65, 9)

============================================================
k = 1
============================================================
Confusion matrix (rows = actual, cols = predicted):
[[15  2  2  0  0  0]
 [ 3 16  0  0  3  1]
 [ 1  0  3  0  0  0]
 [ 0  1  0  5  0  0]
 [ 0  0  0  1  2  0]
 [ 0  0  0  0  1  9]]
Accuracy          : 0.7692
Precision (macro) : 0.7164

============================================================
k = 2
============================================================
Confusion matrix (rows = actual, cols = predicted):
[[16  2  1  0  0  0]
 [ 7 12  1  1  2  0]
 [ 2  1  1  0  0  0]
 [ 0  2  0  4  0  0]
 [ 0  1  0  0  2  0]
 [ 0  1  0  0  1  8]]
Accuracy          : 0.6615
Precision (macro) : 0.6342

============================================================
k = 3
============================================================
```

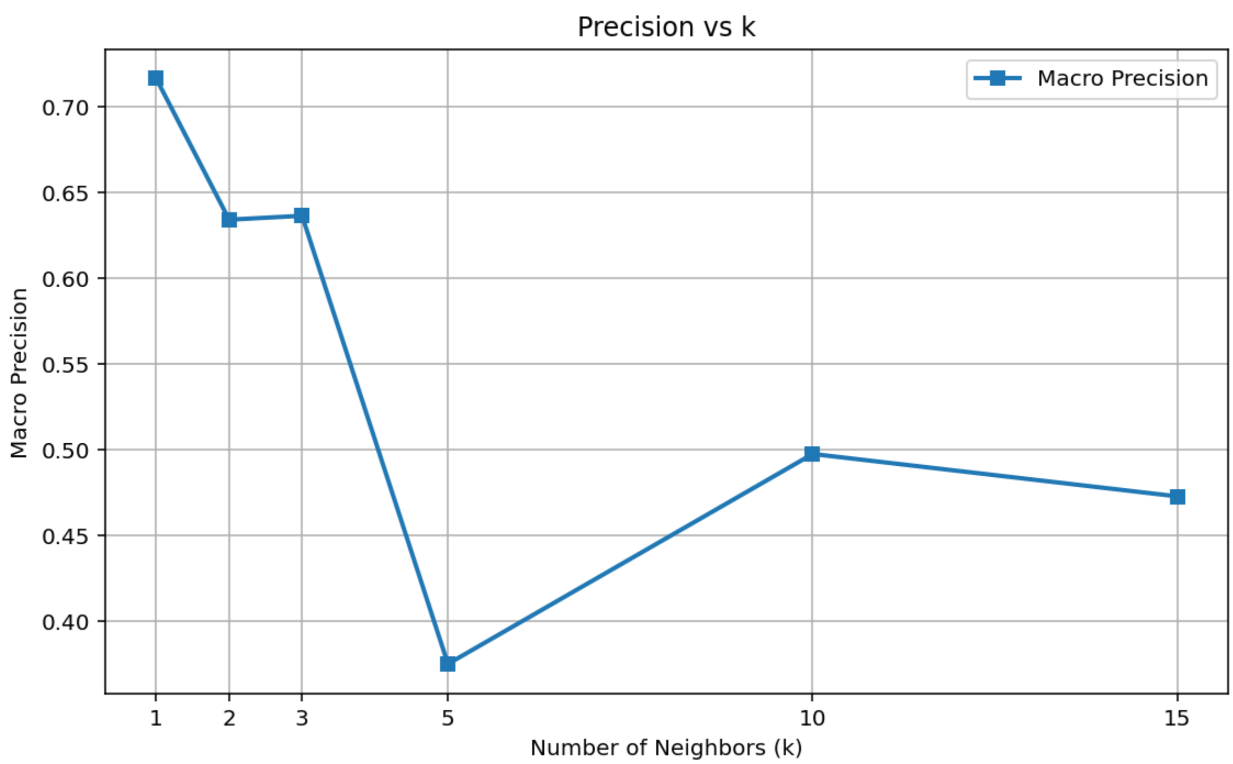
```
Confusion matrix (rows = actual, cols = predicted):
[[13  3  3  0  0  0]
 [ 6 15  0  0  2  0]
 [ 2  0  2  0  0  0]
 [ 1  2  0  3  0  0]
 [ 0  1  0  0  1  1]
 [ 1  0  0  0  1  8]]
Accuracy         : 0.6462
Precision (macro) : 0.6364

==========================================================
k = 5
==========================================================
Confusion matrix (rows = actual, cols = predicted):
[[16  2  1  0  0  0]
 [ 7 13  1  0  2  0]
 [ 4  0  0  0  0  0]
 [ 1  4  0  0  0  1]
 [ 0  1  0  0  1  1]
 [ 1  0  0  0  1  8]]
Accuracy         : 0.5846
Precision (macro) : 0.3753

==========================================================
k = 10
==========================================================
Confusion matrix (rows = actual, cols = predicted):
[[17  2  0  0  0  0]
 [11 11  0  0  1  0]
 [ 4  0  0  0  0  0]
 [ 1  3  0  2  0  0]
 [ 1  1  0  0  0  1]
 [ 1  1  0  0  0  8]]
Accuracy         : 0.5846
Precision (macro) : 0.4976

==========================================================
k = 15
==========================================================
Confusion matrix (rows = actual, cols = predicted):
[[17  2  0  0  0  0]
 [10 11  1  0  0  1]
 [ 3  1  0  0  0  0]
 [ 0  4  0  2  0  0]
```

[1 0 0 0 0 2]
[1 1 0 0 0 8]]
Accuracy : 0.5846
Precision (macro) : 0.4729

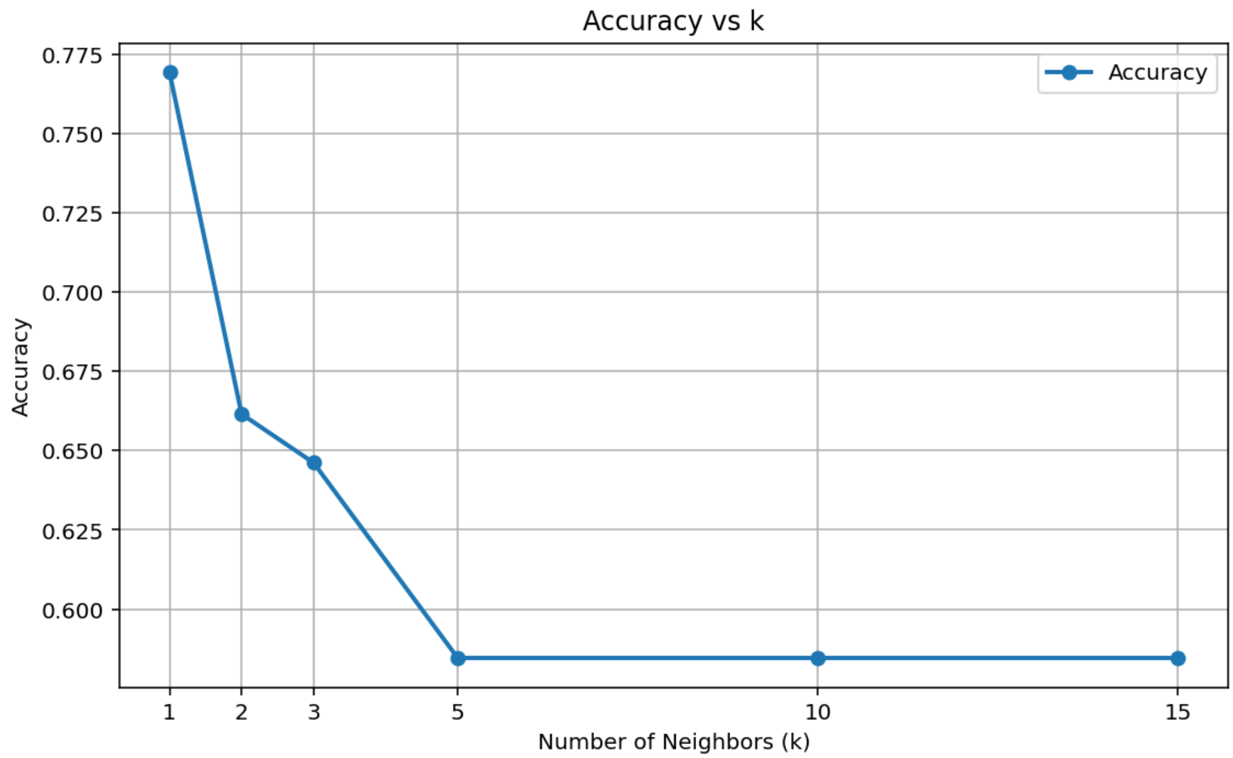
==
SUMMARY
==

k	accuracy	precision
1	0.7692	0.7164
2	0.6615	0.6342
3	0.6462	0.6364
5	0.5846	0.3753
10	0.5846	0.4976
15	0.5846	0.4729

Best accuracy : k = 1 (accuracy = 0.7692)
Best precision: k = 1 (precision = 0.7164)



The figure illustrates the effect of the hyperparameter k on the macro precision of the K-Nearest Neighbors classifier. The highest macro precision (71.64%) was achieved when k = 1, indicating that the nearest-neighbor approach provided the best classification performance for the Glass Identification dataset. As the value of k increased, macro precision generally decreased due to the averaging effect of considering more neighboring samples, which reduced the classifier's ability to distinguish between overlapping classes.



The figure illustrates the effect of the hyperparameter k on the classification accuracy of the K-Nearest Neighbors (KNN) classifier. The highest accuracy (76.92%) was achieved when k = 1, indicating that considering the single nearest neighbor provided the best classification performance on the Glass Identification dataset. As the value of k increased, the classification accuracy gradually decreased and stabilized at approximately 58.46% for k = 5, 10, and 15. This behavior suggests that using a larger neighborhood introduced additional misclassifications due to overlapping class distributions in the dataset.